Vietnam National University of HCMC
International University
School of Computer Science and Engineering

Program of MIT
Course: Advanced Computer Graphics (IT516)

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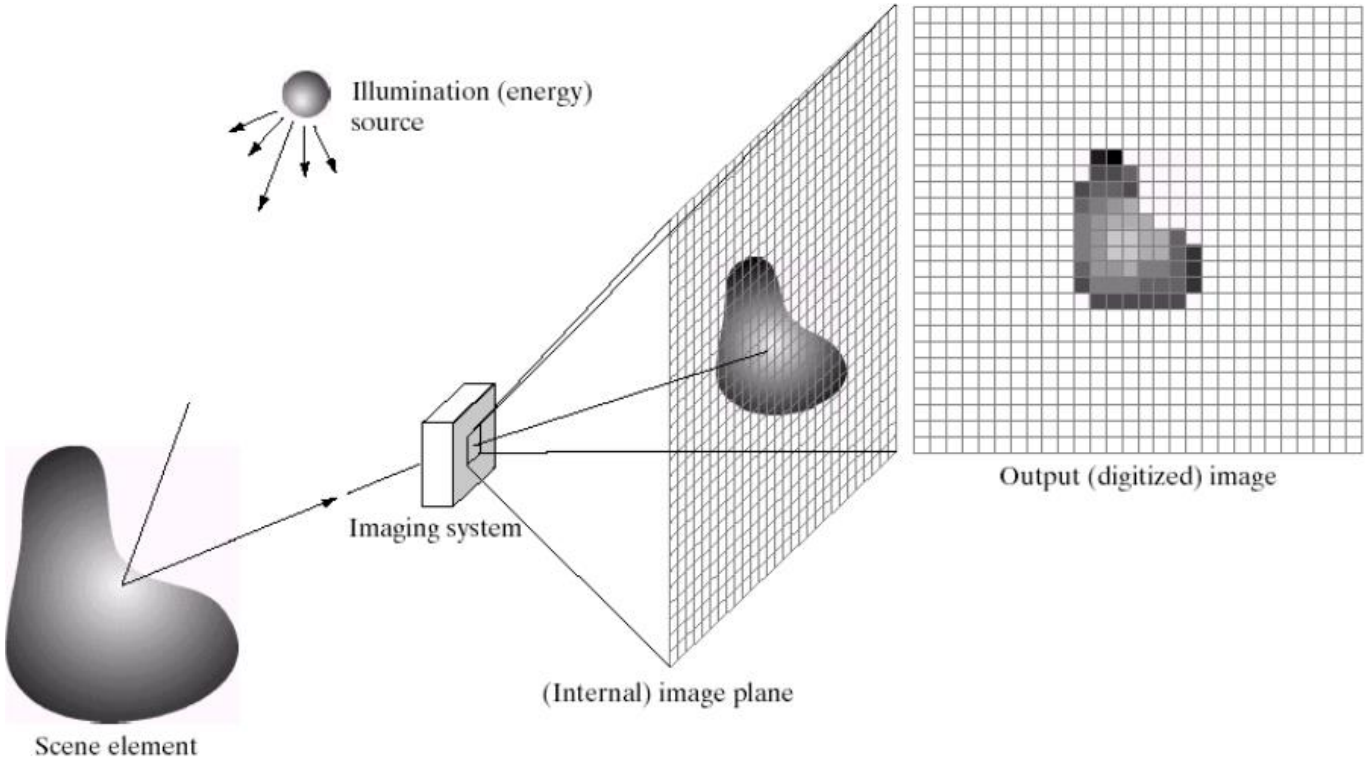
Image representation & processing

- Introduction to digital image
- Image representation
- Methods for image processing
  - Pixels, Intensity value, Grey level
  - Histogram, color processing
  - Noise removing
  - Fourier transformation
  - Segmentation
  - Contour (boundary)

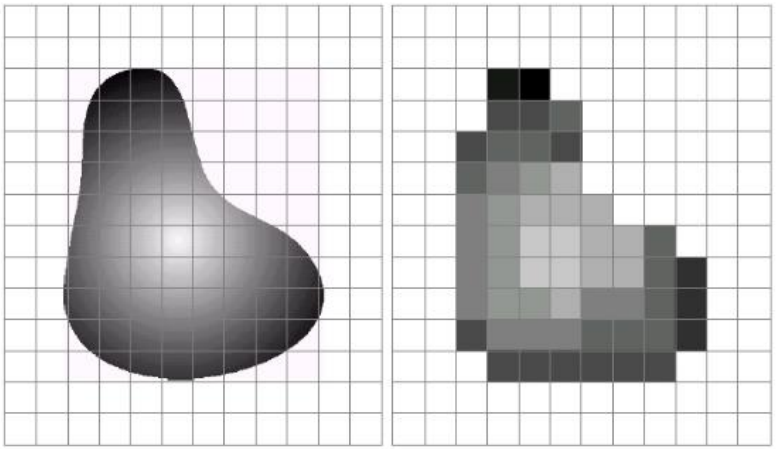
What is a digital image?

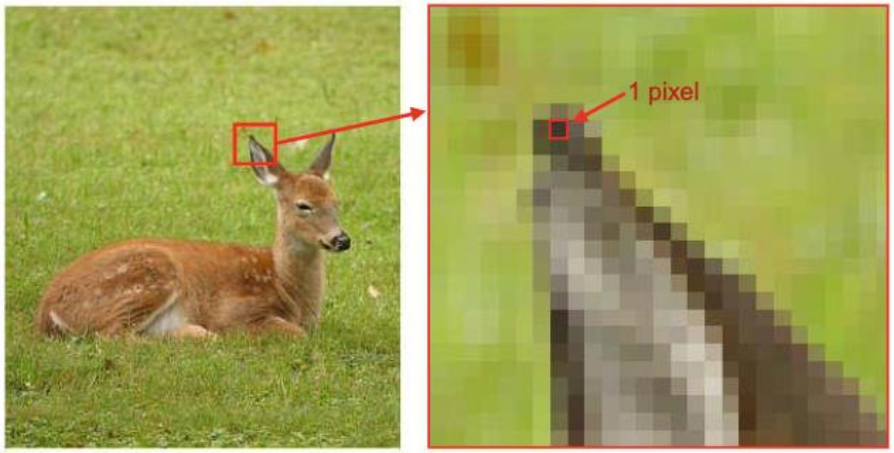
- A digital image is a representation of a 2D image as a finite set of digital values, called picture elements or pixels



What is a digital image?

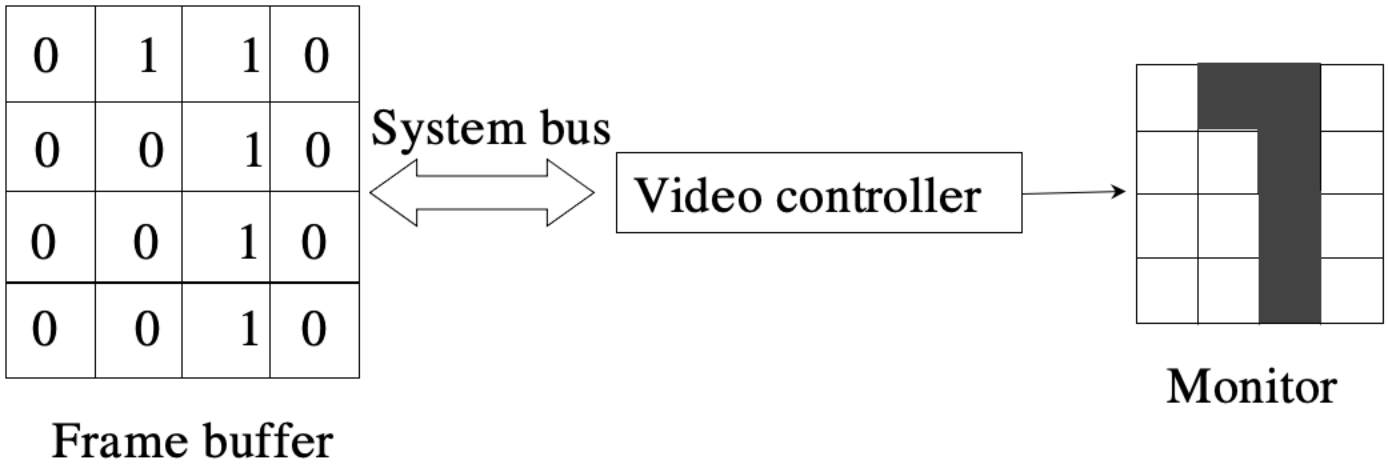
- Pixel values typically represent gray levels, colors, heights, opacities etc.,
- Remember *digitization* implies that a digital image is an *approximation* of a real scene.





What is a digital image?

- A bit on the display frame buffers corresponds with a pixel (a display point) on the monitor

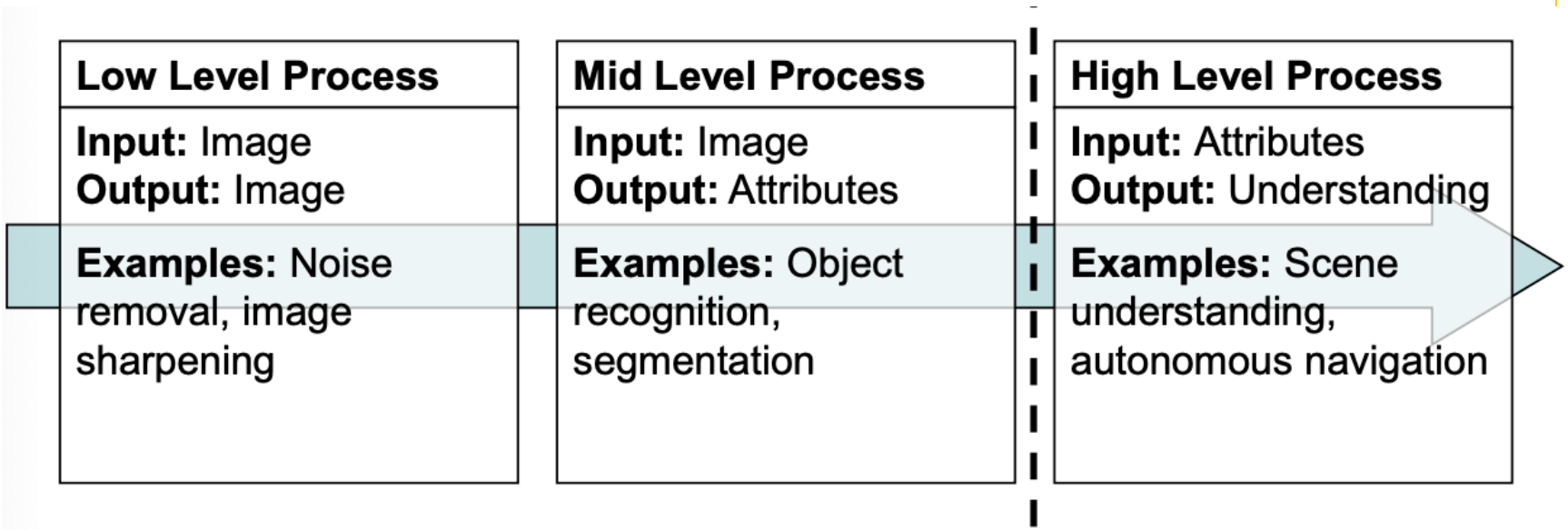


What is digital image processing?

- Digital image processing focuses on two major tasks
  - Improvement of pictorial information for human interpretation
  - Processing of image data for storage, transmission and representation for autonomous machine perception
- Some argument about where image processing ends and fields such as image analysis and computer vision start

What is digital image processing?

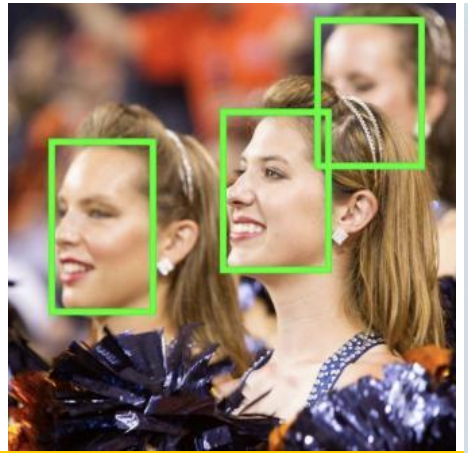
- The process from image processing to computer vision can be broken up into low-level, mid-level and high-level processes:



Example: from an image (or a set of images) to a 3D object

Application of image processing

- Try to make human computer interfaces more natural
  - Face recognition
  - Gesture recognition
- Does anyone remember the user interface from “Minority Report”?
- These tasks can be extremely difficult



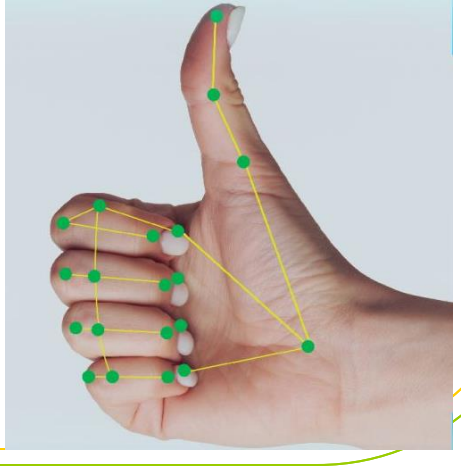


Image representation

- After an image has been segmented into regions; the resulting pixels is usually represented and described in suitable form for further computer processing.
- Representing regions relates into two choices:
  1) External characteristics (boundary)
  2) Internal characteristics (pixels)
- Task (1) is making data useful to computer
- Task (2) is to describe the region of representation

Image representation

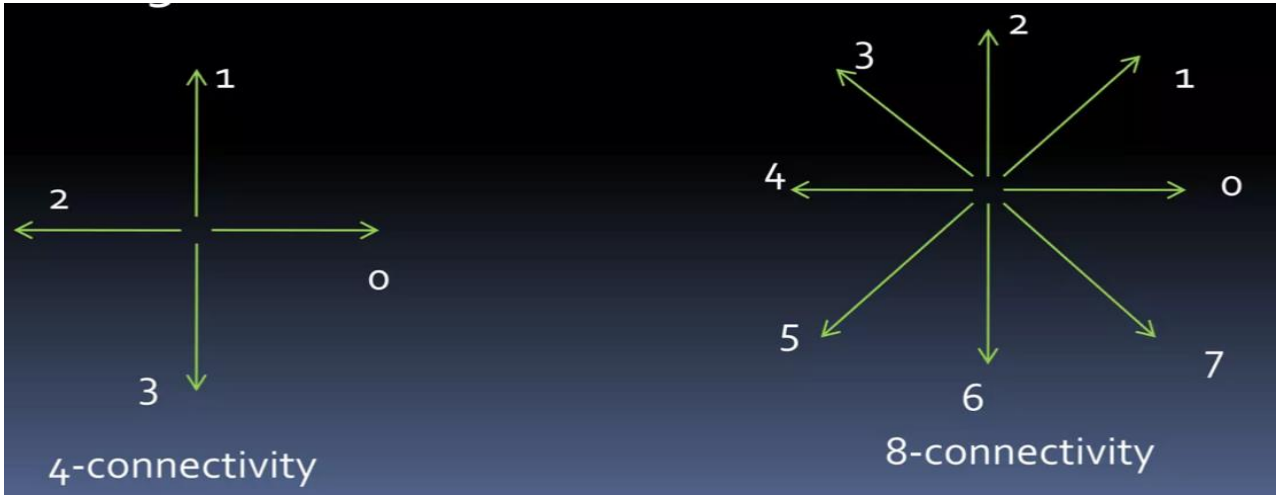
- External representation is chosen when primary focus on shape characteristics.
- Internal representation is chosen when primary focus on region properties like color and texture characteristics.
- In some cases, it is possible to choose both characteristics.
- Features selected as descriptors should be insensitive to changes in size, translation & rotation.

Image representation

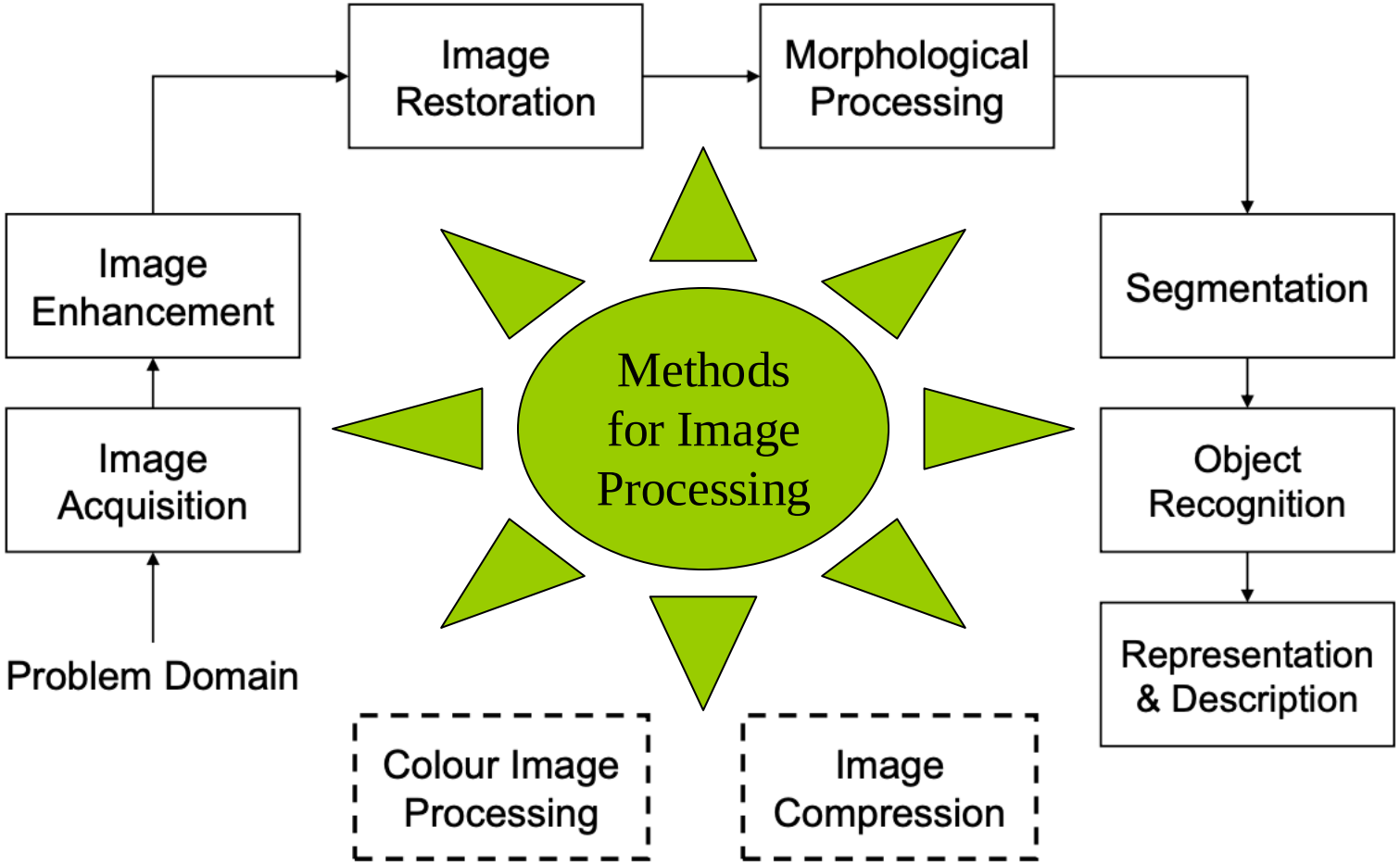
- It deals with compaction of segmented data into representations that facilitate the computation of descriptors.
- Technique of Representation:
  - a. Chain codes
  - b. Polygonal Approximations
    - Minimum perimeter polygone
    - Merging technique
    - Splitting technique
  - c. Signatures
  - d. Boundary Segments
  - e. Skeletons

Image representation

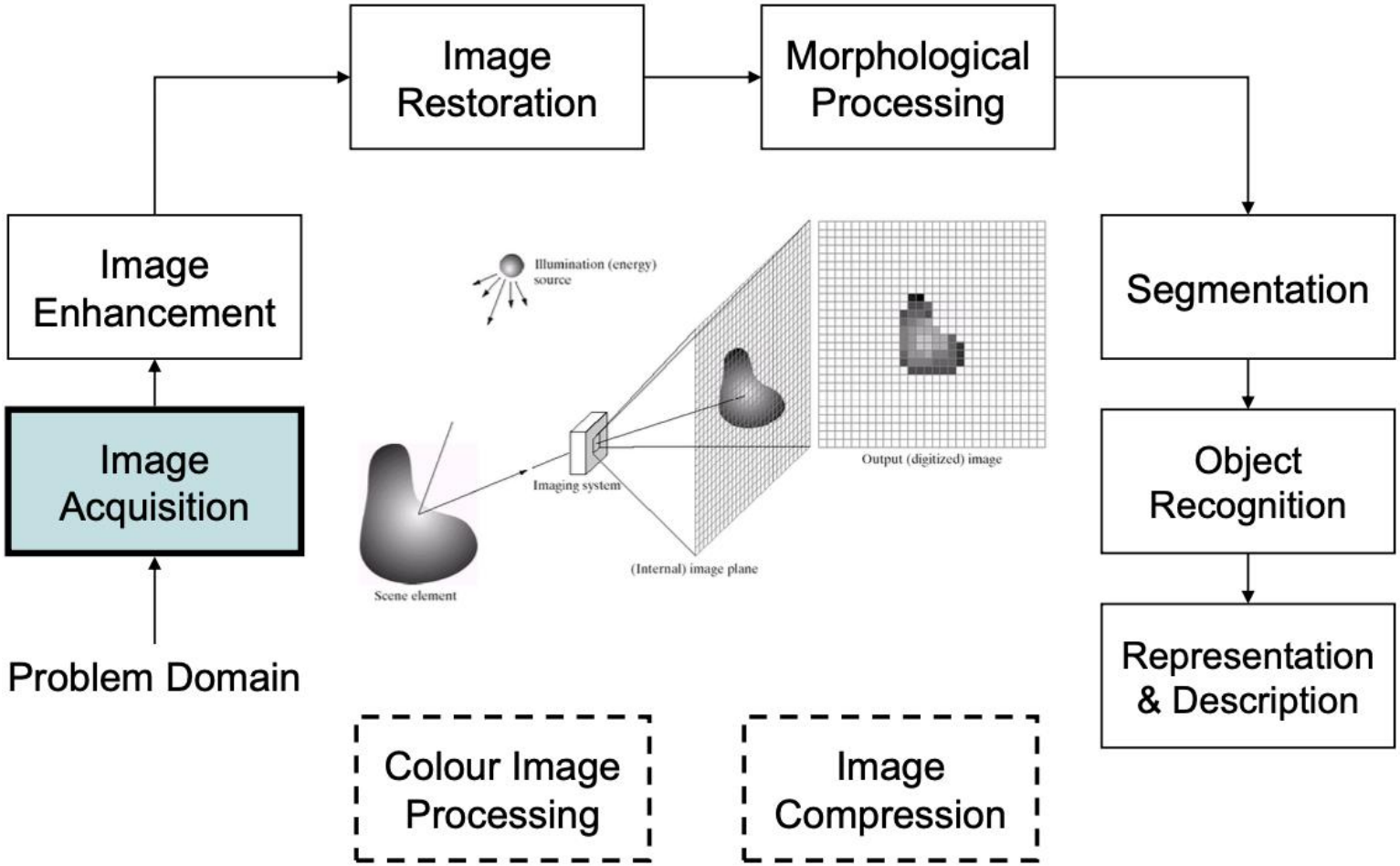
- They are used to represent a boundary by a connected sequence of straight line segments of specified length & direction.
- Typically this representation is based on 4- or 8- connectivity of segments.
- The direction of each segments is coded by using a numbering scheme.



Methods for image processing



Methods for image processing



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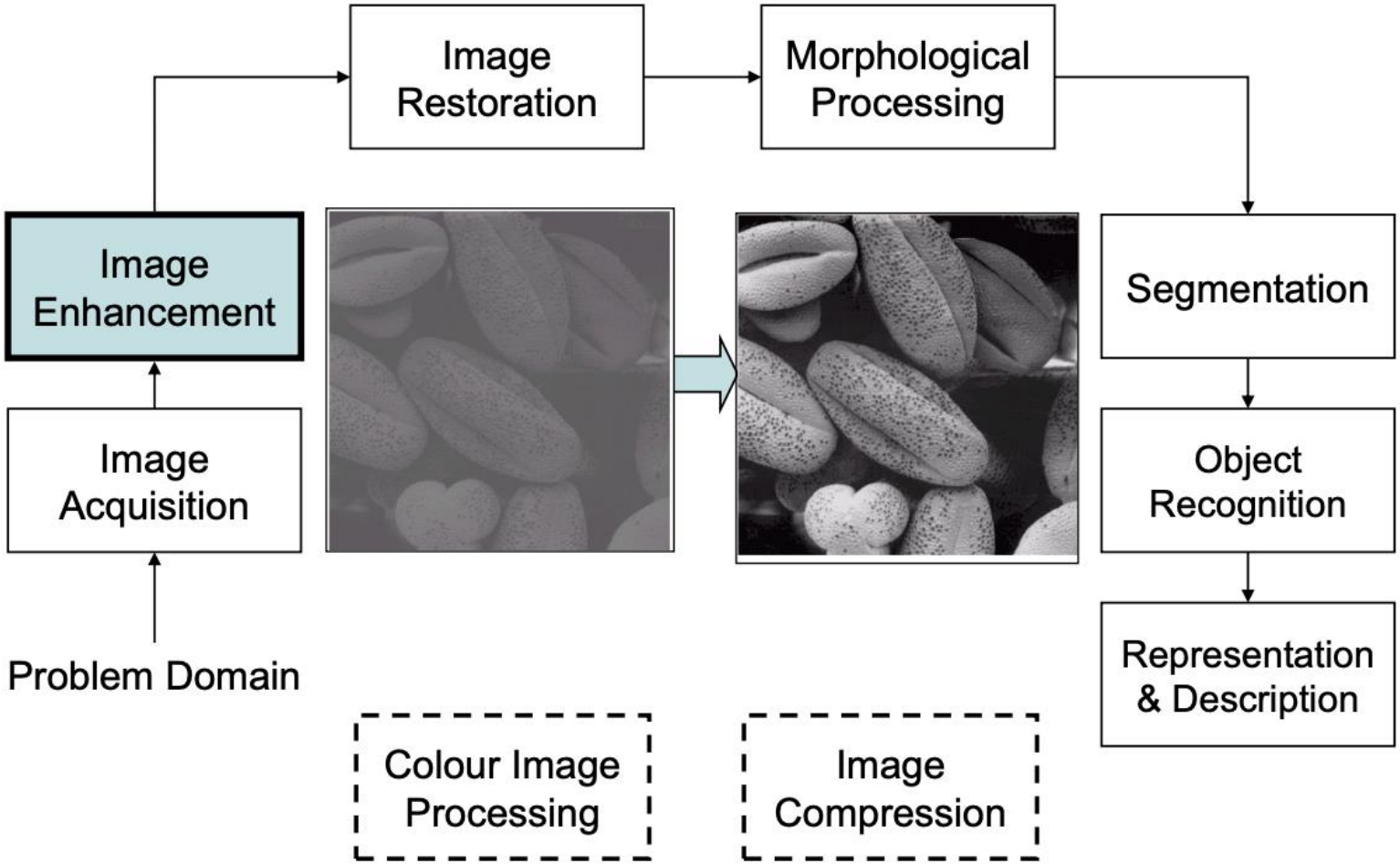
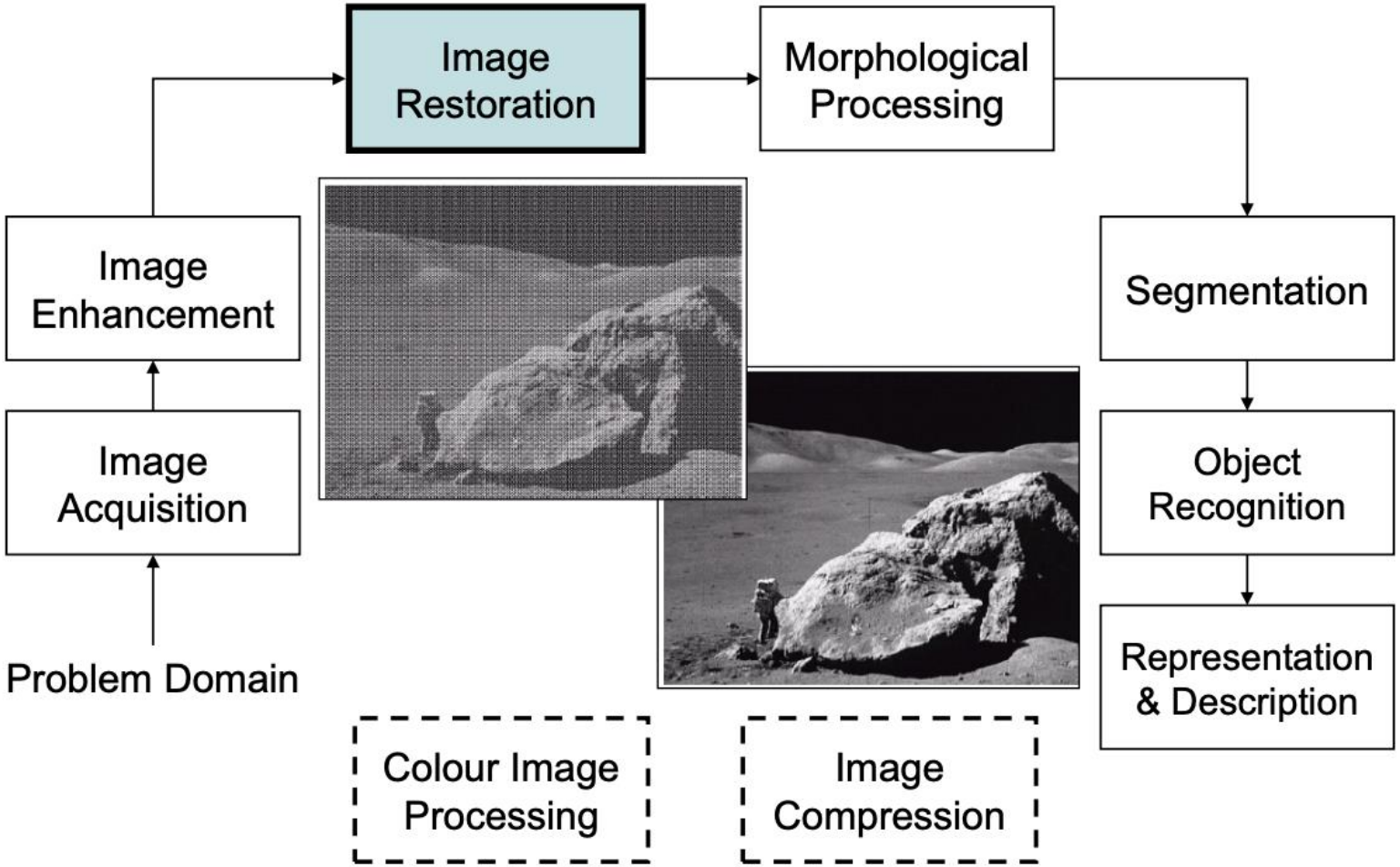


Methods for image processing



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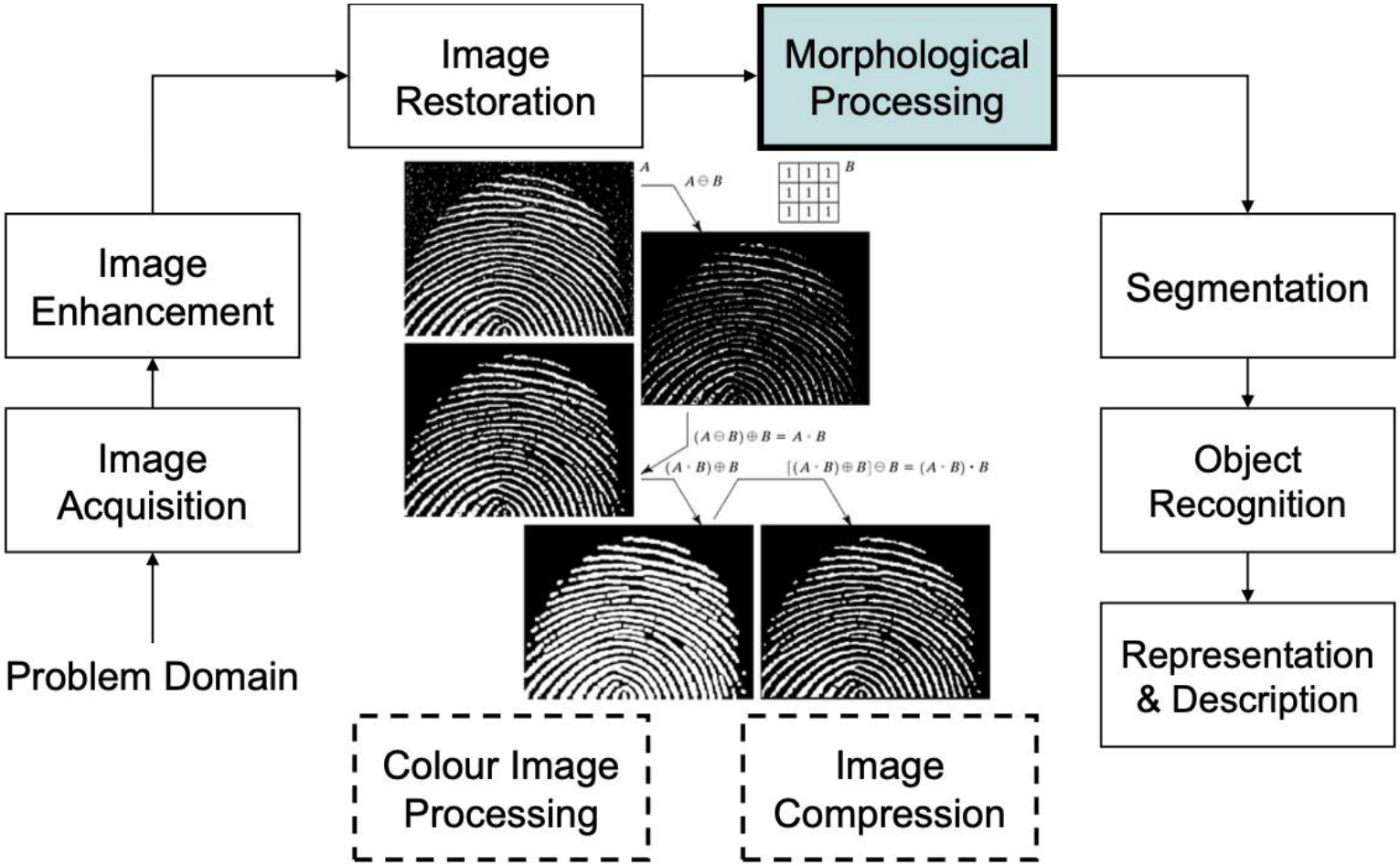

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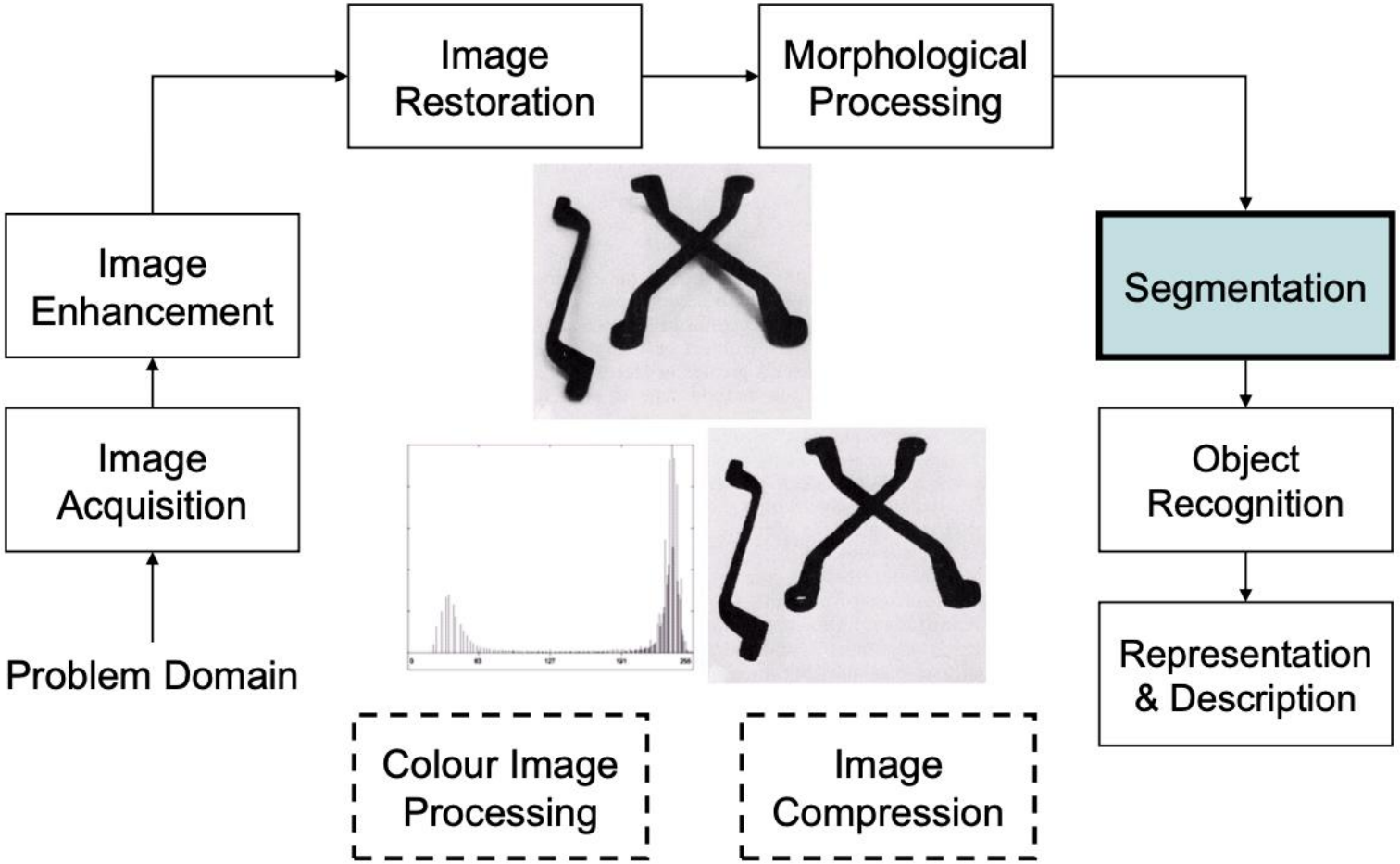

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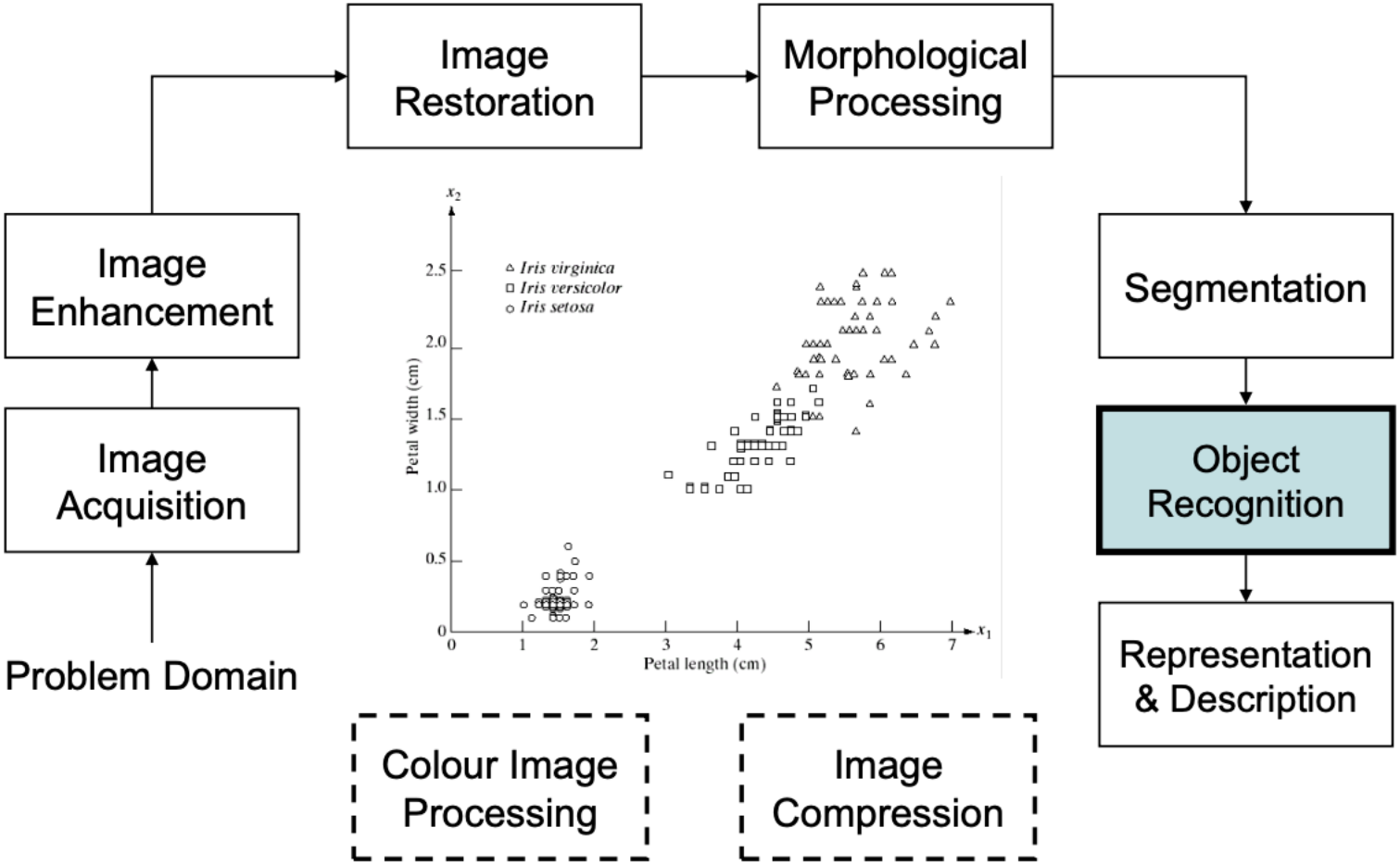
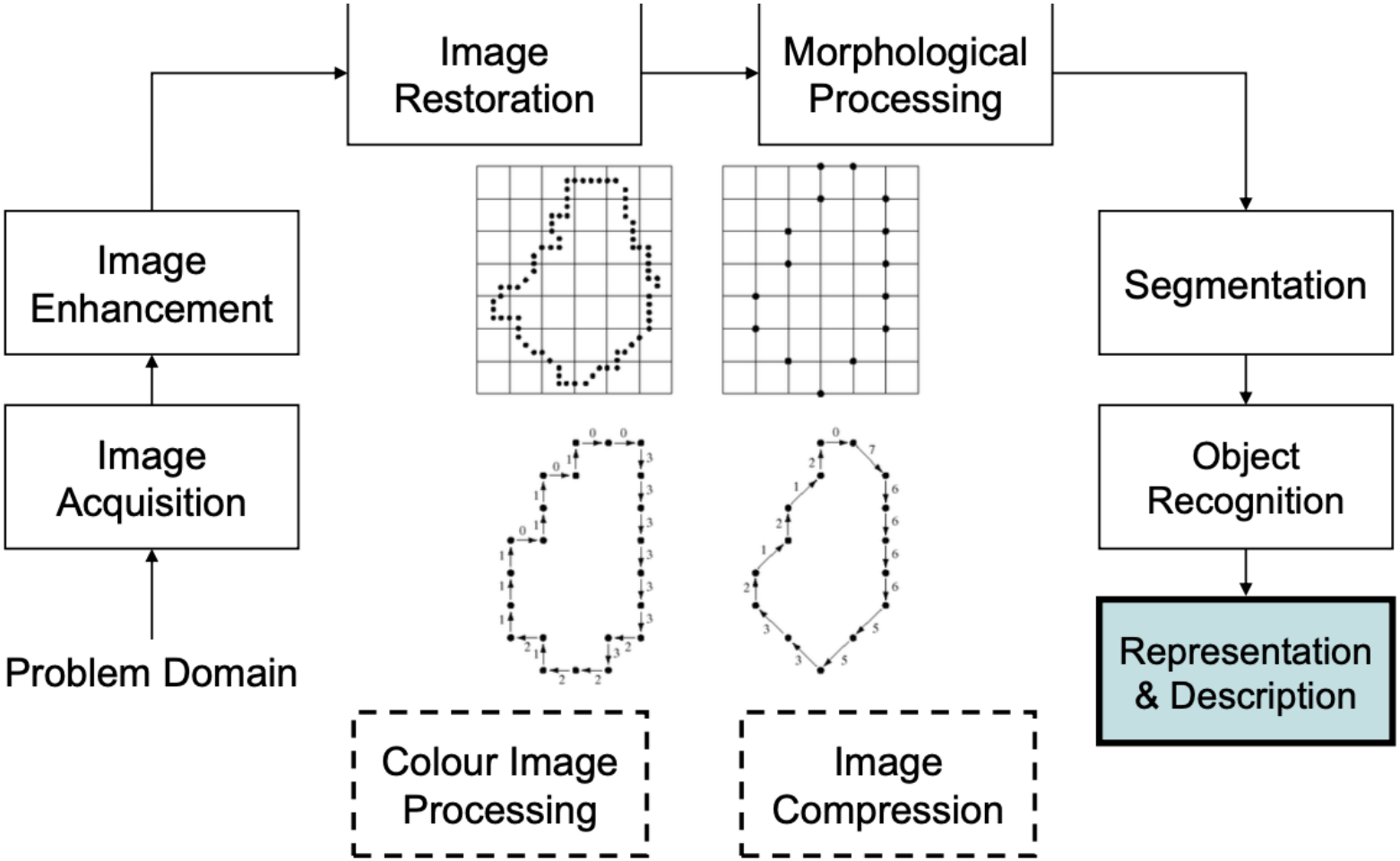
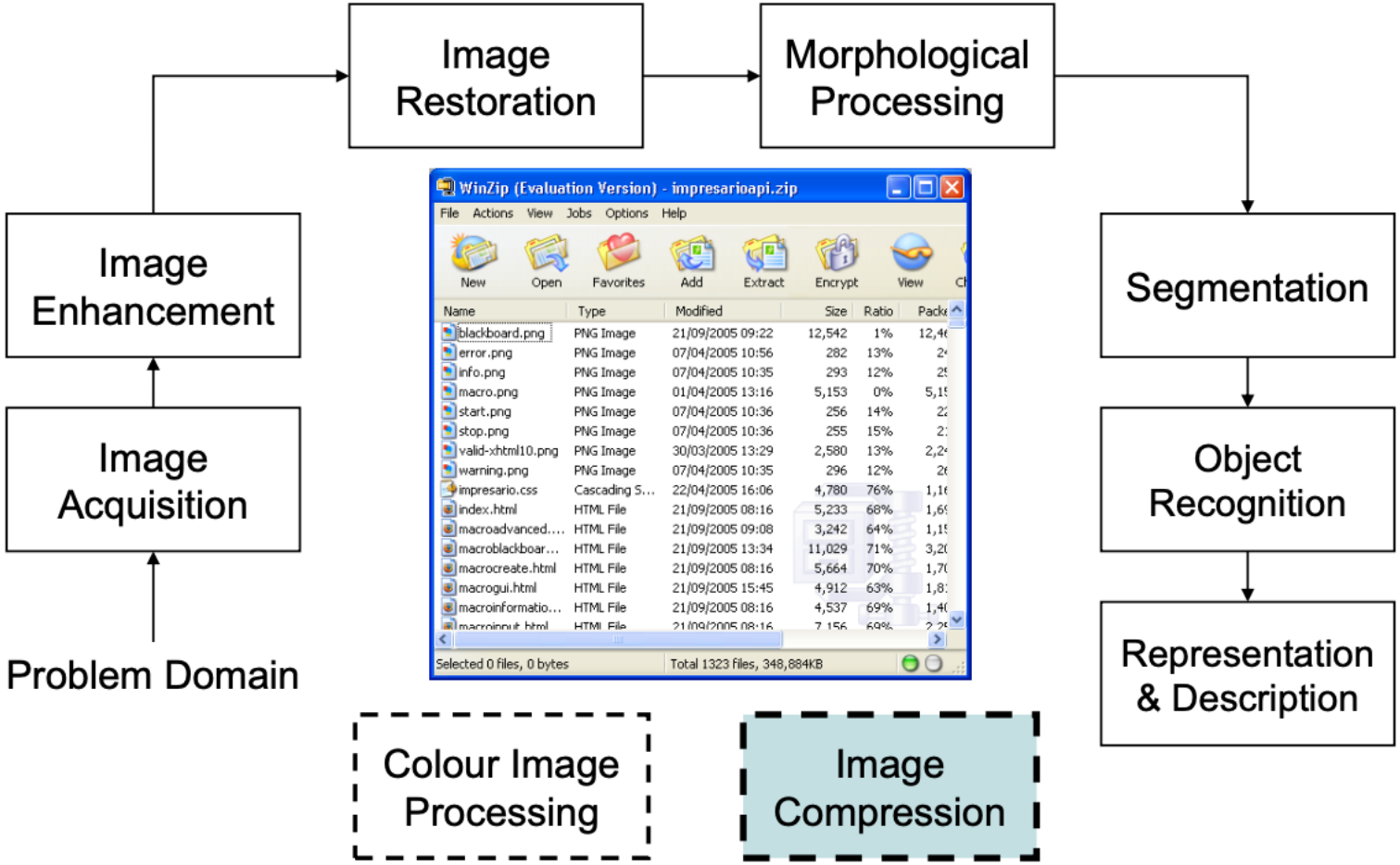


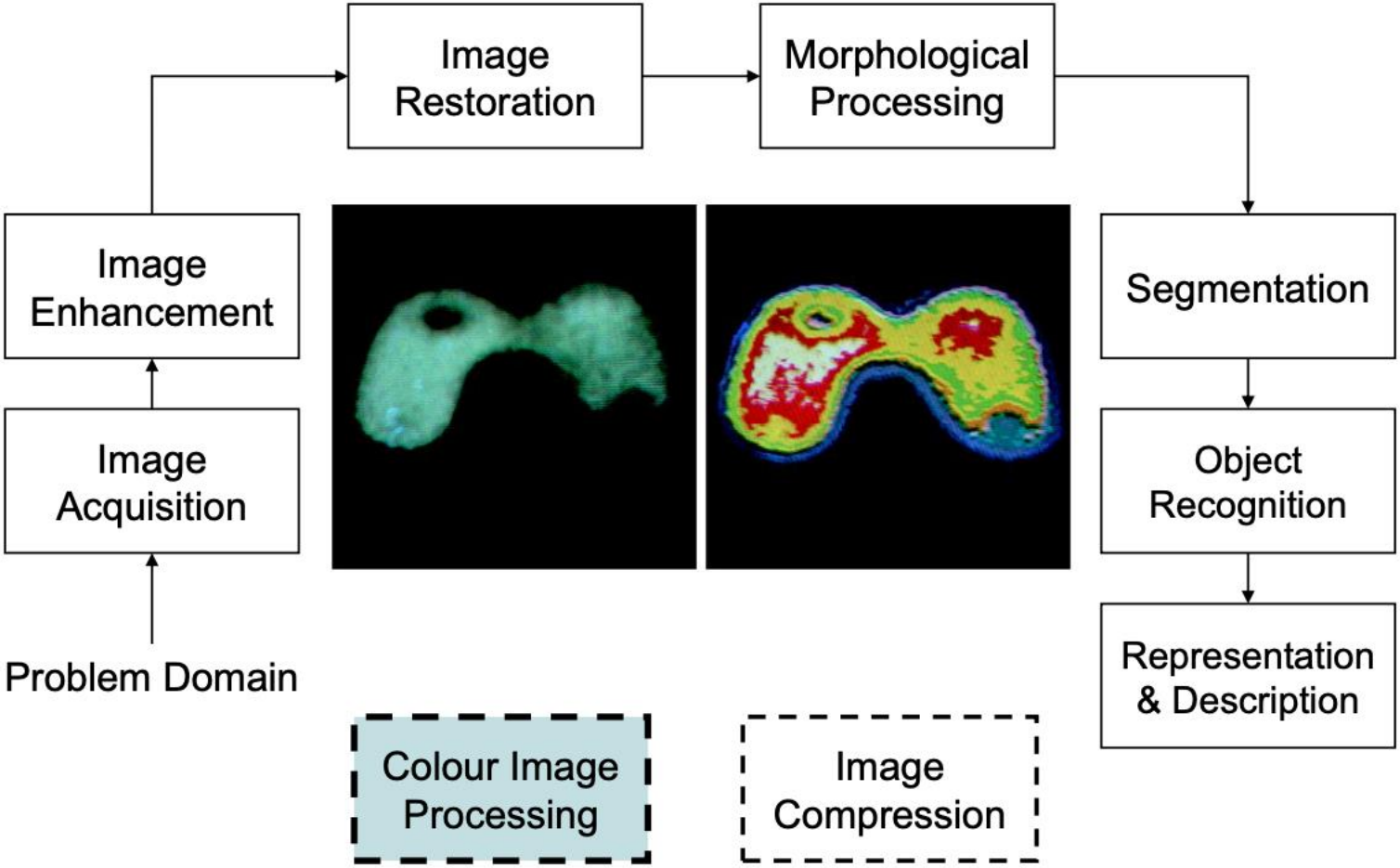
Methods for image processing



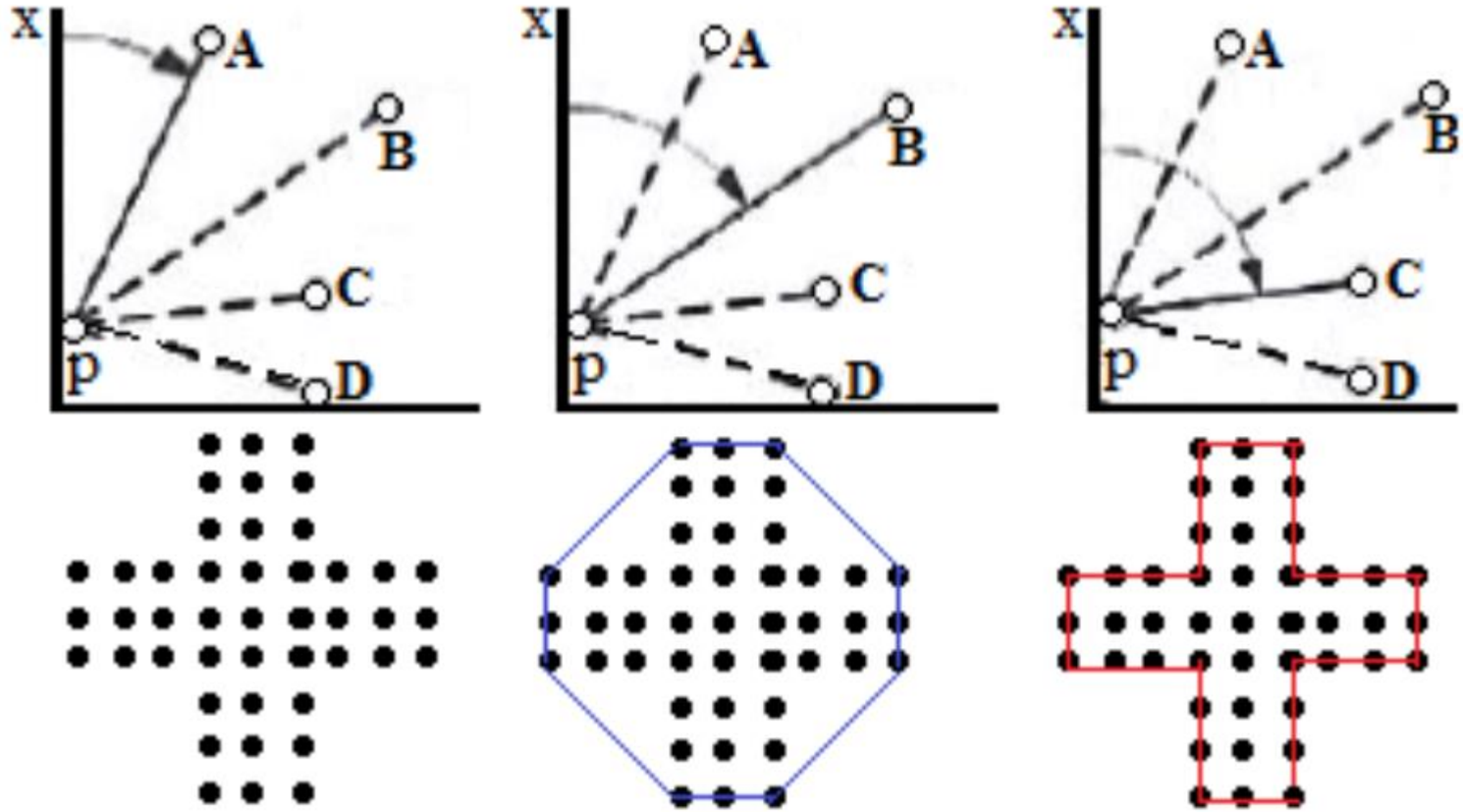
Methods for image processing



Methods for image processing



Algorithms for boundary processing



a) Raw points b) Convex hull c) Boundary

Algorithms for boundary processing

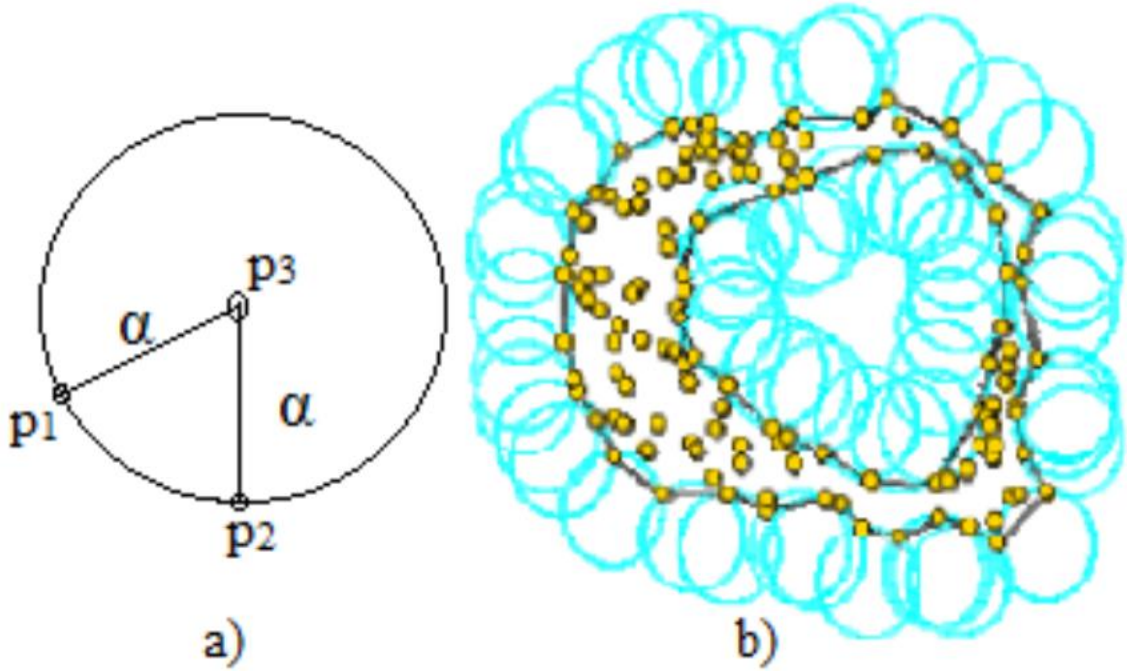


Figure 3.2: Method to extract a boundary of Shen Wei [Wei08]

Algorithms for boundary processing

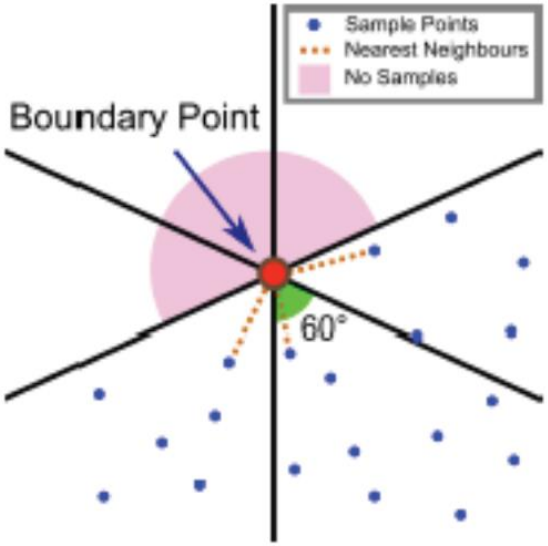


Figure 3.4: Method to extract the boundary points of Kalogerakis [KNSS09]

Algorithms for boundary processing

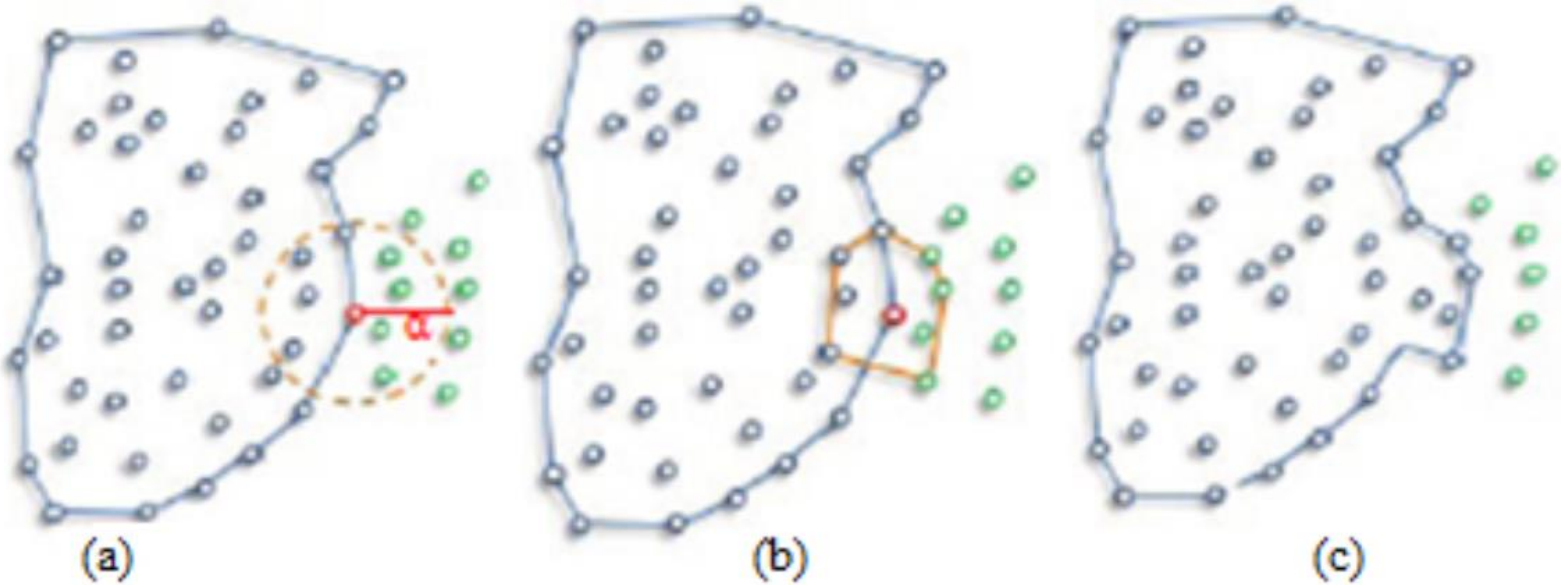
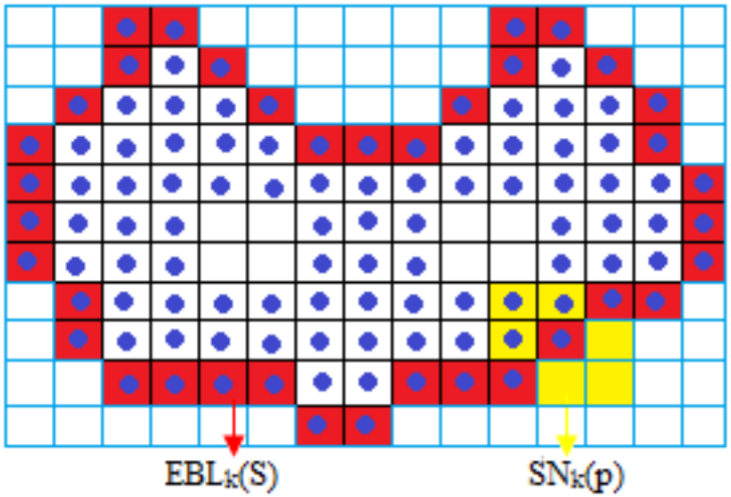


Figure 3.5: Method to extract a boundary of Sait [SKM11]

Algorithms for boundary processing



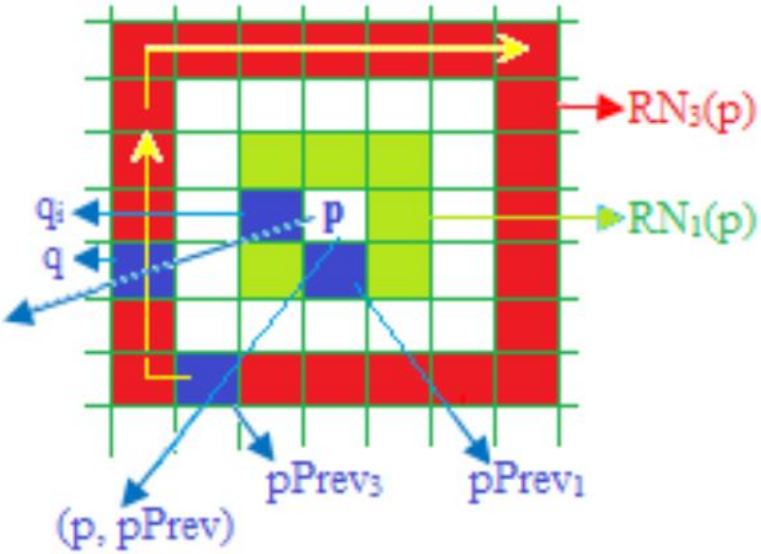


Figure 3.10: An exterior boundary with $k = 1$

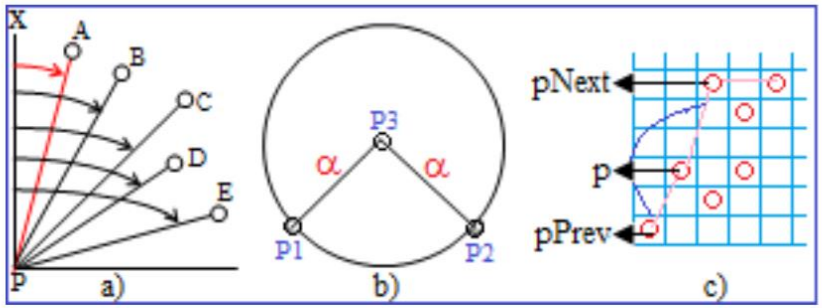


Figure 3.17: Comparison of computing strategies between the methods: Sampath (a), ShenWei (b) and Our method (c).

Example for boundary processing

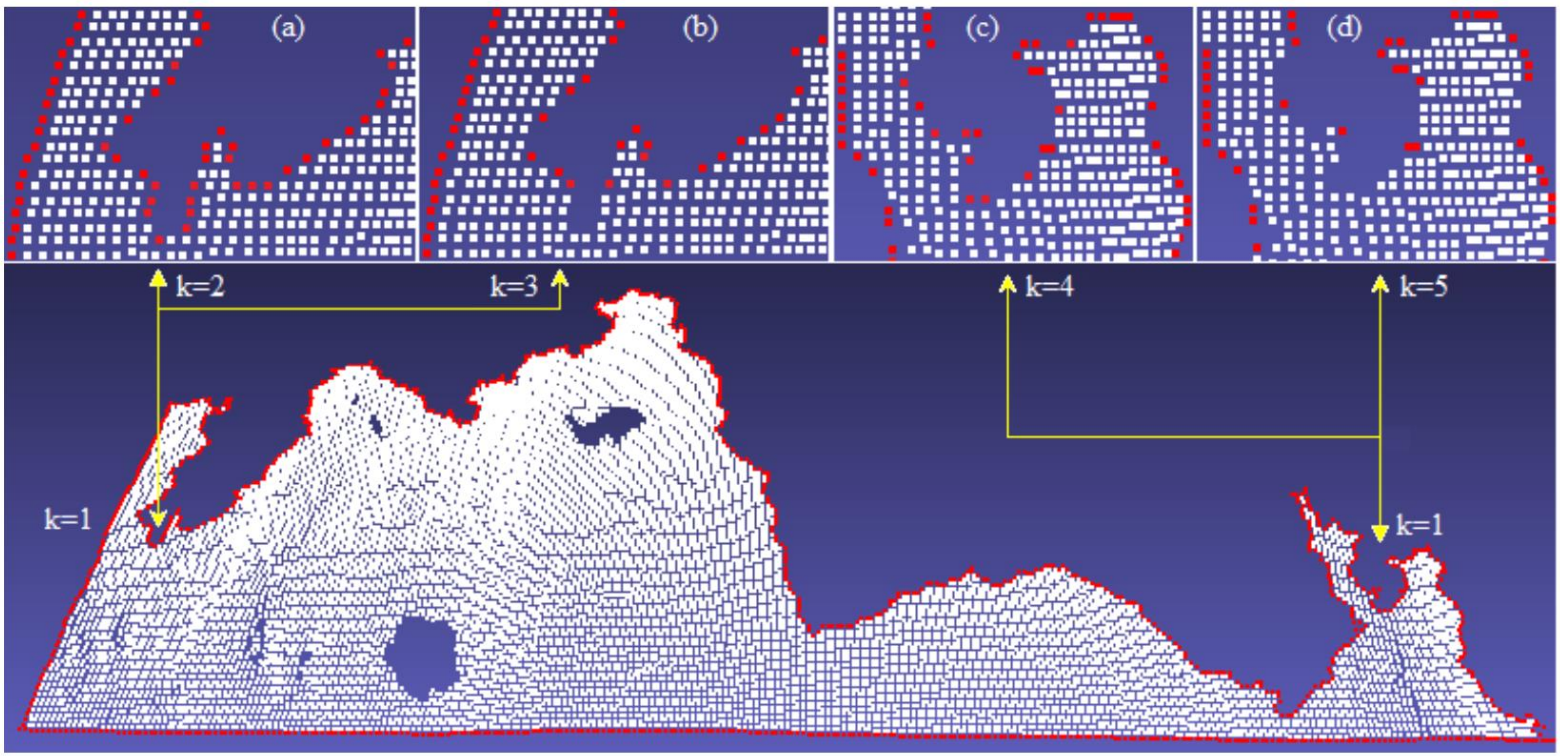
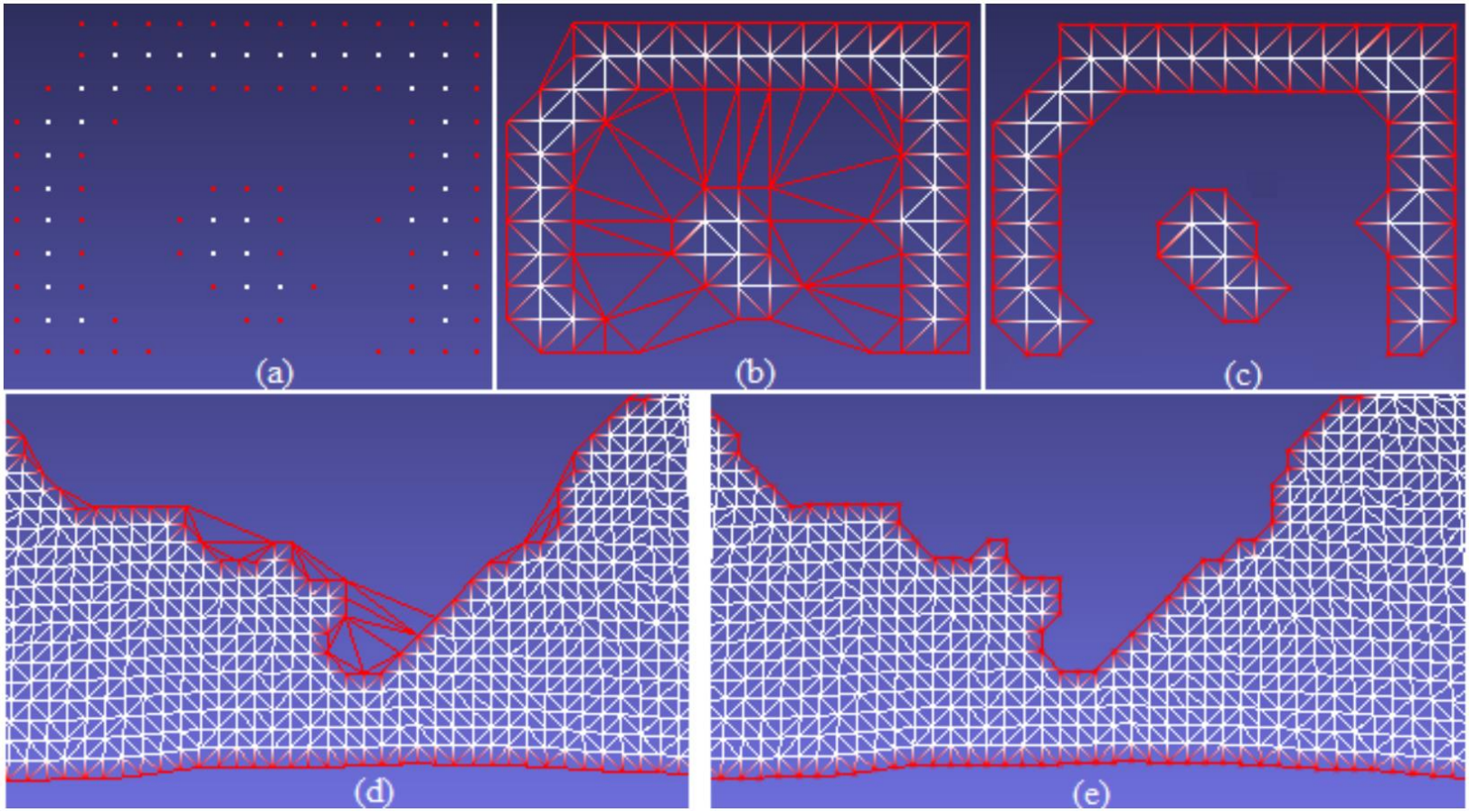


Figure 3.27: The shape of the boundary has been changed on the concave parts after extracting the boundary using many values of k. The resolution of the boundary is inversely proportional to the value of k: the smaller the value of k, the higher the resolution of the boundary; and in contrast, the more increasing values of k, the more rough boundaries we obtain.

Algorithms for boundary processing



Fourier transformation algorithm

- The Fourier transform (FT) is a mathematical method to transform signals from time domain to frequency, applied in image processing to obtain image with higher quality.
- FT is representation of an image as a sum of complex exponentials of varying magnitudes, frequencies, and phases.
- FT plays a critical role in a broad range of image processing applications, including enhancement, analysis, restoration, and compression.
- Detail: see DFT (Discrete Fourier Transform)

  (reference: Vietnamese book)